

OSSP (Operating Systems And Systems Programming) - 25CS2104E

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int source, destination;
    char buffer[100];
    int bytes;

    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        printf("Error opening source file\n");
        return 1;
    }

    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0)
    {
        printf("Error opening destination file\n");
        close(source);
        return 1;
    }

    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");
}
```

```
    return 0;  
}
```

```
aruntej@LAPTOP-DGRT8C5G:~$ echo "Hello, this is a sample  
aruntej@LAPTOP-DGRT8C5G:~$ cat source.txt  
Hello, this is a sample file.  
aruntej@LAPTOP-DGRT8C5G:~$ nano experiment-2.c  
aruntej@LAPTOP-DGRT8C5G:~$ gcc experiment-2.c -o experiment-2  
aruntej@LAPTOP-DGRT8C5G:~$ ./expreiment-2  
-bash: ./expreiment-2: No such file or directory  
aruntej@LAPTOP-DGRT8C5G:~$ ./experiment-2  
File copied successfully.  
aruntej@LAPTOP-DGRT8C5G:~$ cat destination.txt  
Hello, this is a sample file.
```

```

+++ exited with 0 +++
aruntej@LAPTOP-DGRT8C5G:~$ strace -e trace=openat,read,write,close cat sample
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 4096) = 16
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 4096) = 16
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 4096) = 16
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\250\2\0\0\0\0"..., 4096) = 16
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 4096) = 16
close(3) = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "/dev/sdd / ext4 rw,relatime,disc"..., 1024) = 1024
read(3, " tmpfs rw,nosuid,nodev,mode=755 "..., 1024) = 1024
read(3, ",upperdir=/gpu_lib/rw/upper,work"..., 1024) = 1024
read(3, "th=C:\\;uid=1000;gid=1000;symlink"..., 1024) = 299
read(3, "", 1024) = 0
close(3) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "5639fd74f000-5639fd841000 r--p 0"..., 1024) = 1024
read(3, "0 08:30 13647"..., 1024) = 1024
read(3, "00 08:30 13650"..., 1024) = 1024
read(3, " /usr/lib/x86_64-lin"..., 1024) = 1024
read(3, "0 13644 /us"..., 1024) = 566
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"..., 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
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close(5) = 0
close(4) = 0
close(3) = 0
+++ exited with 0 +++
aruntej@LAPTOP-DGRT8C5G:~$ |

```